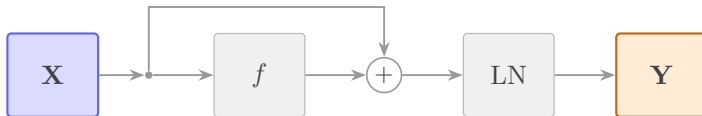


Where layer-norm sits: pre-norm vs post-norm

Pre-norm



Post-norm



sublayer $f \in \{\text{MHA}, \text{FFN}\}$, each wrapped identically

pre-norm $\mathbf{Y} = \mathbf{X} + f(\text{LN}(\mathbf{X}))$ post-norm $\mathbf{Y} = \text{LN}(\mathbf{X} + f(\mathbf{X}))$

only LN's position moves — deciding whether the skip (identity) path reaches the next layer untouched